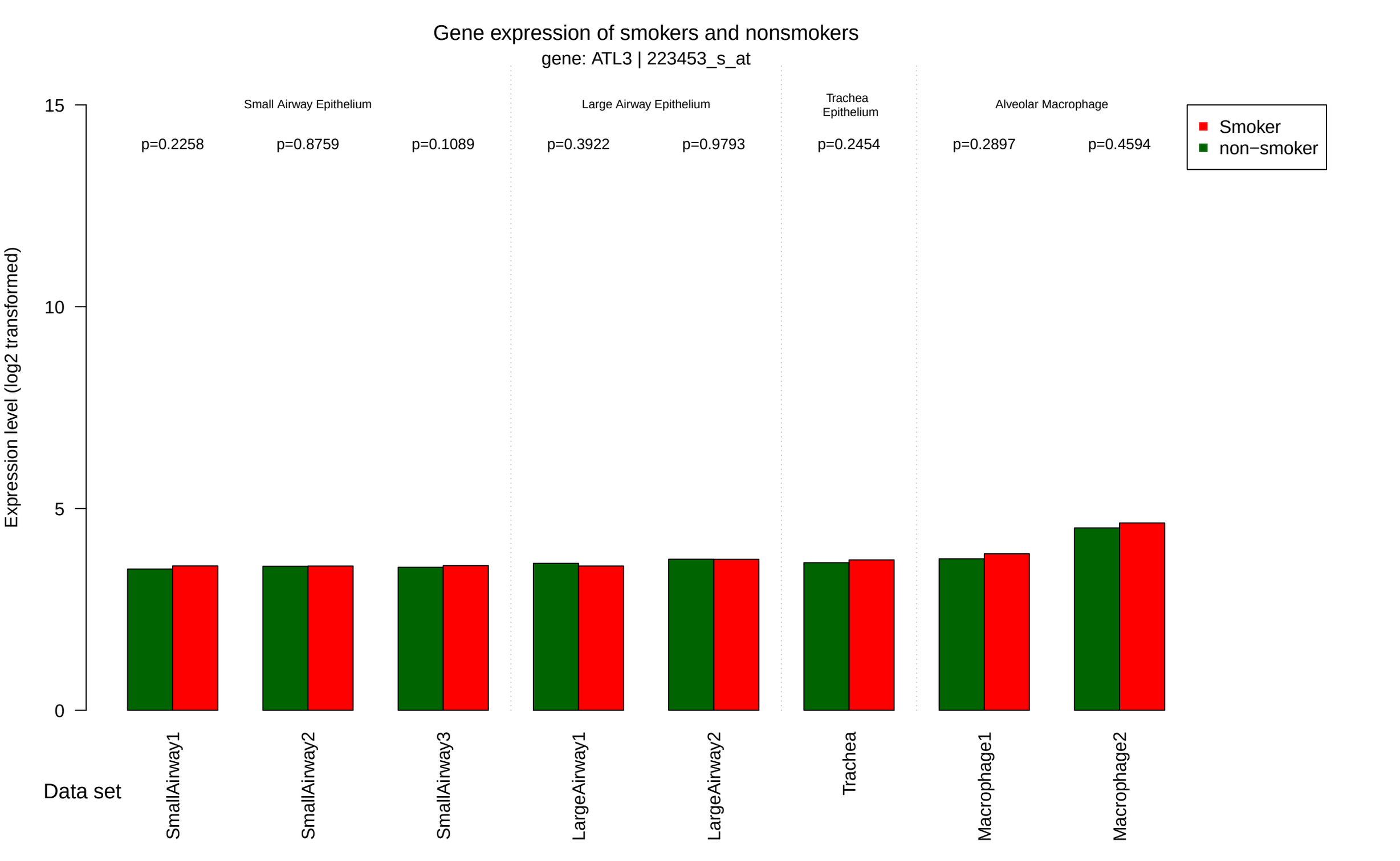
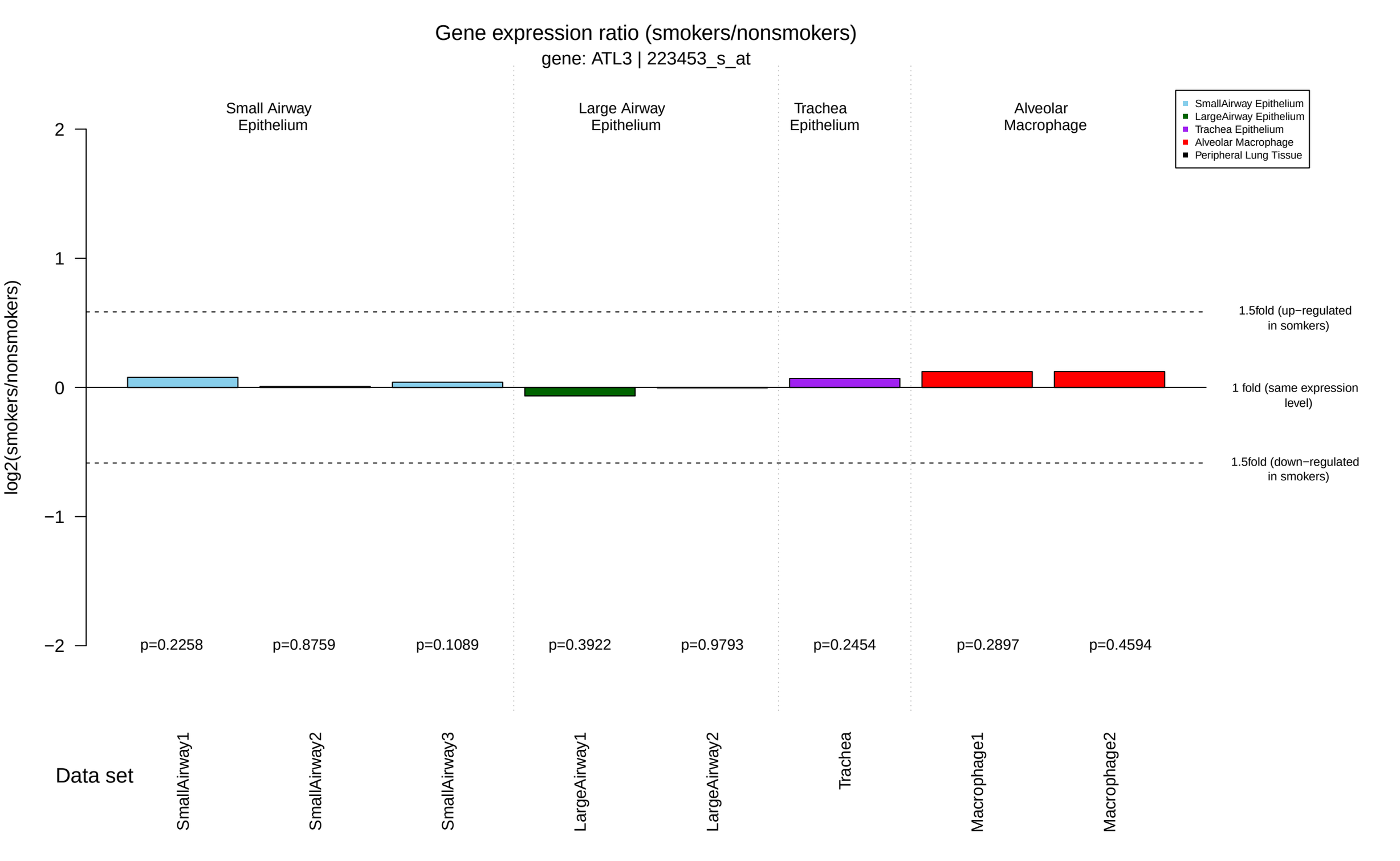
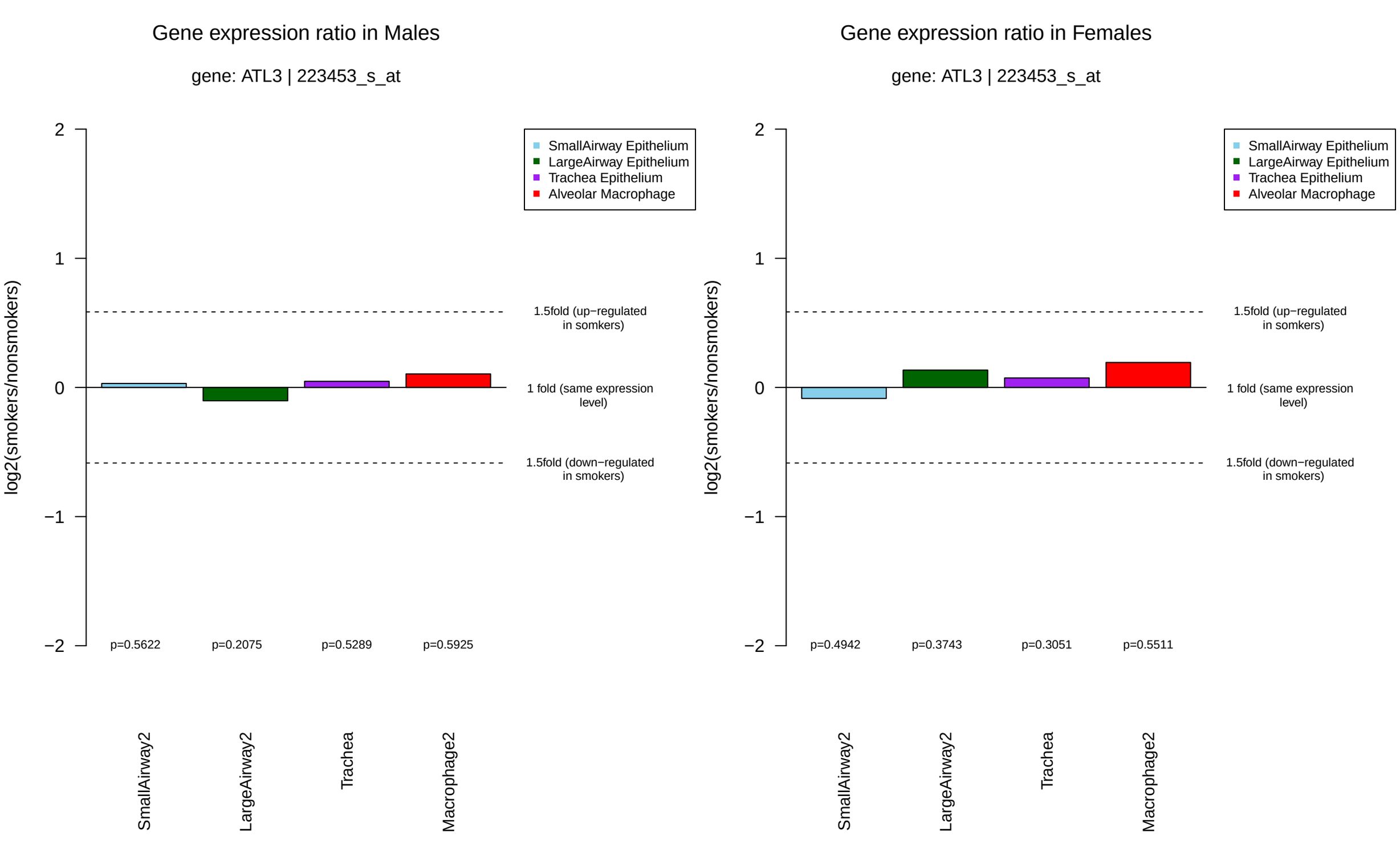
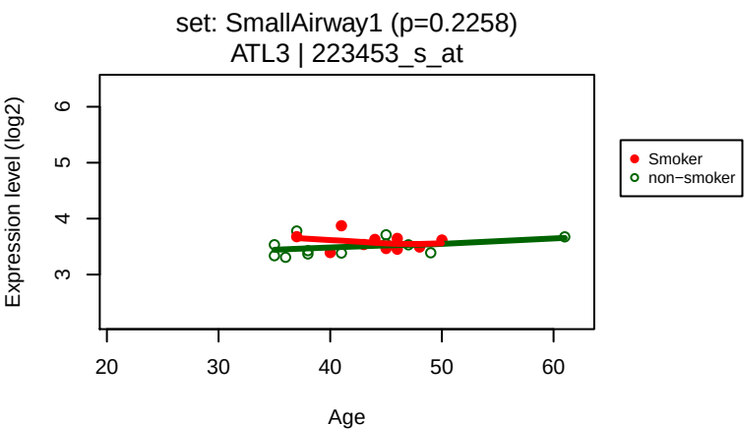
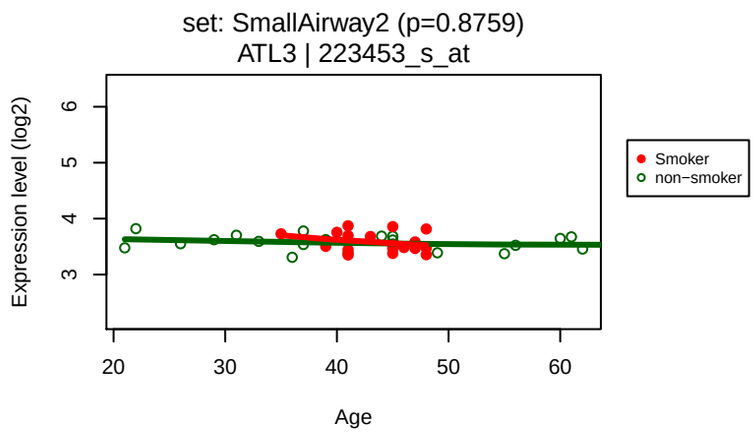
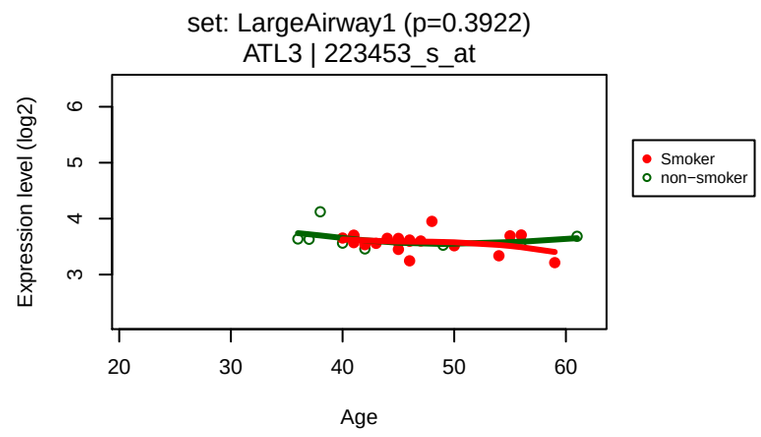

Gene expression of smokers and nonsmokers
gene: ATL3 | 223453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2258
p=0.8759
p=0.1089
p=0.3922
p=0.9793
p=0.2454
p=0.2897
p=0.4594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

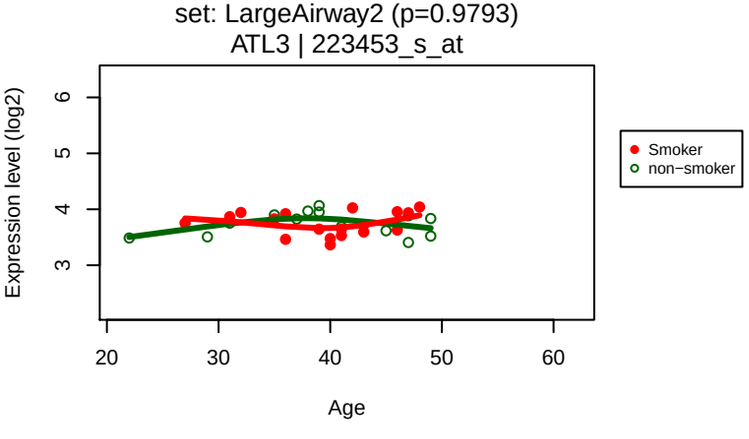

Gene expression ratio (smokers/nonsmokers)
gene: ATL3 | 223453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2258
p=0.8759
p=0.1089
p=0.3922
p=0.9793
p=0.2454
p=0.2897
p=0.4594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

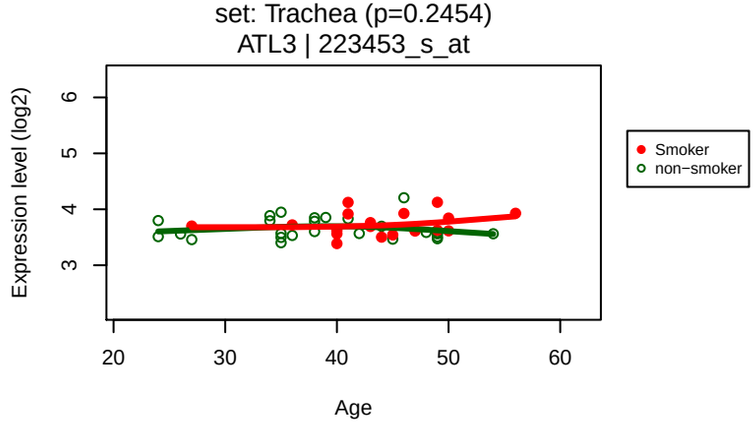

Gene expression ratio in Males
gene: ATL3 | 223453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5622
p=0.2075
p=0.5289
p=0.5925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATL3 | 223453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4942
p=0.3743
p=0.3051
p=0.5511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

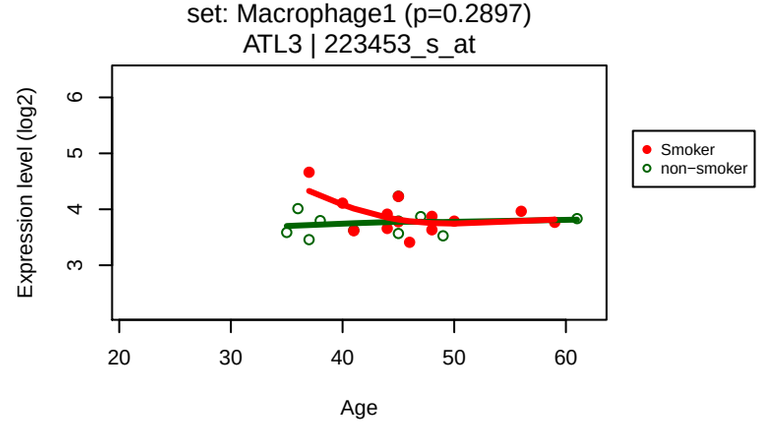

set: SmallAirway1 (p=0.2258)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker

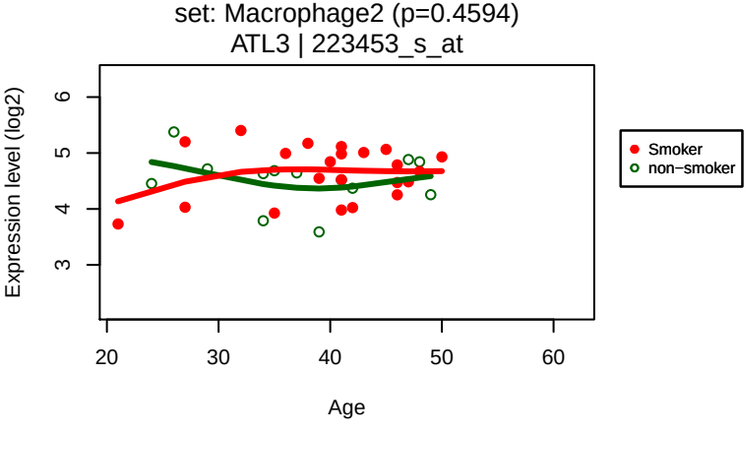

set: SmallAirway2 (p=0.8759)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker

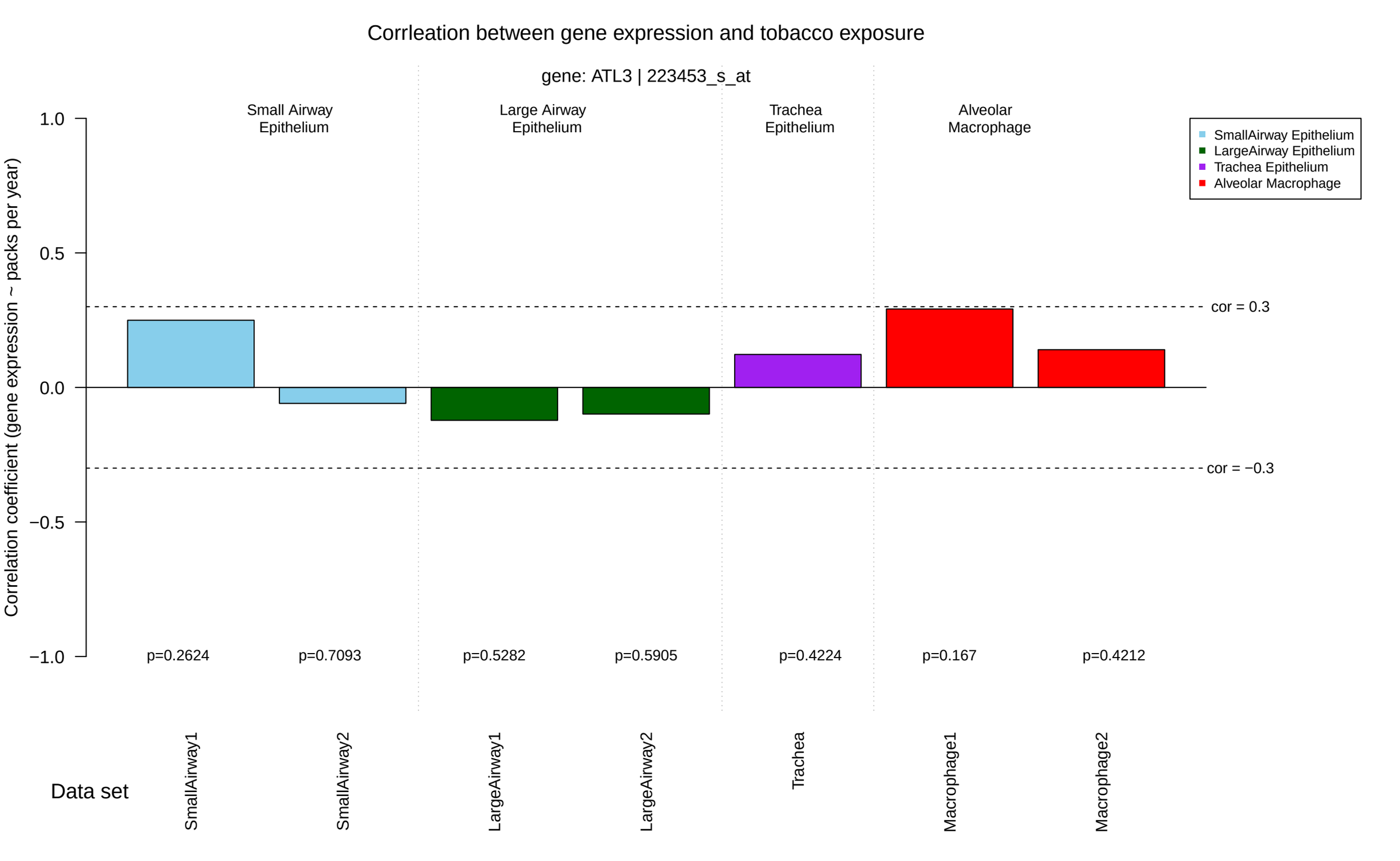

set: LargeAirway1 (p=0.3922)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9793)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2454)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2897)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4594)
ATL3 | 223453_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATL3 | 223453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2624
p=0.7093
p=0.5282
p=0.5905
p=0.4224
p=0.167
p=0.4212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2